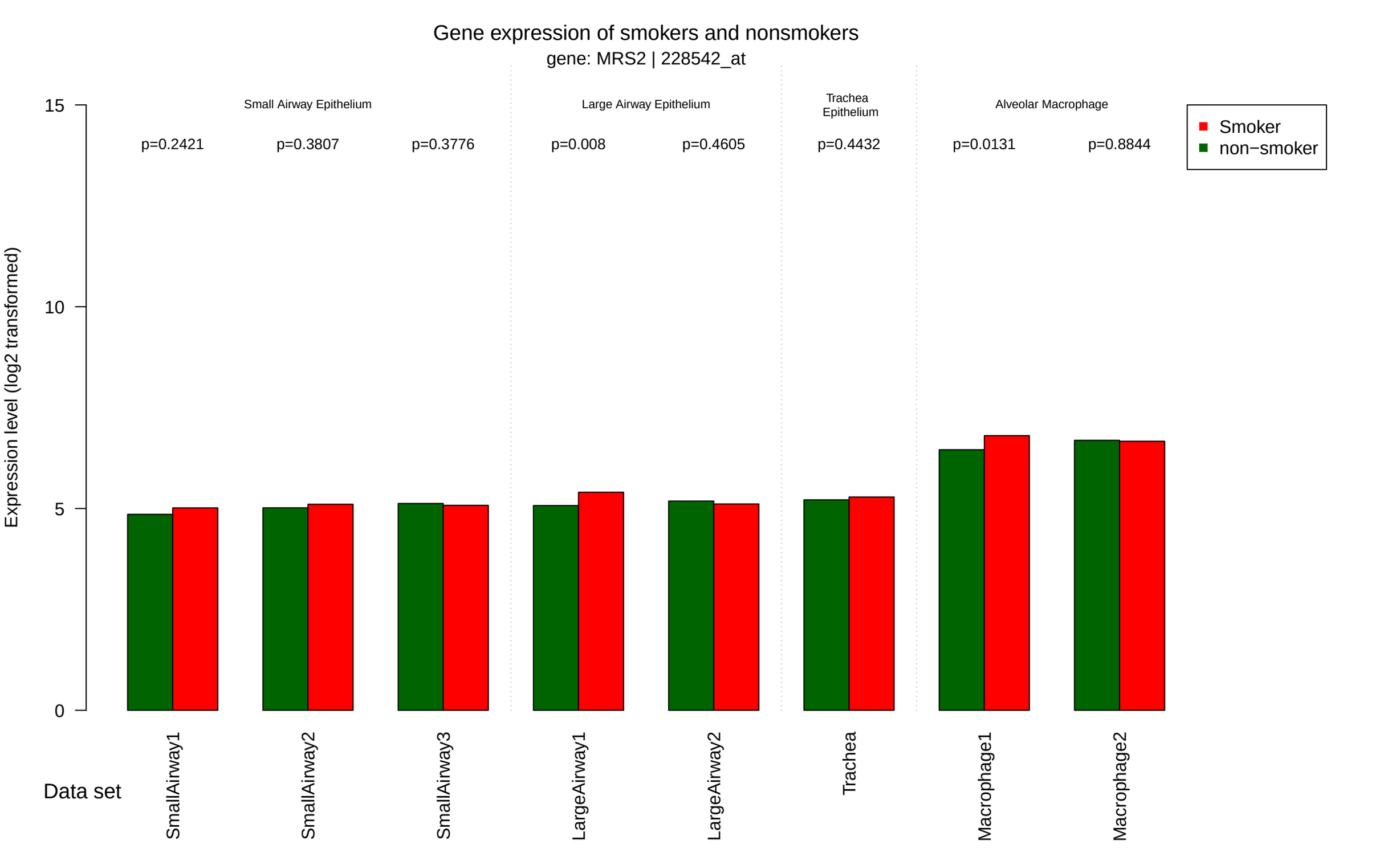
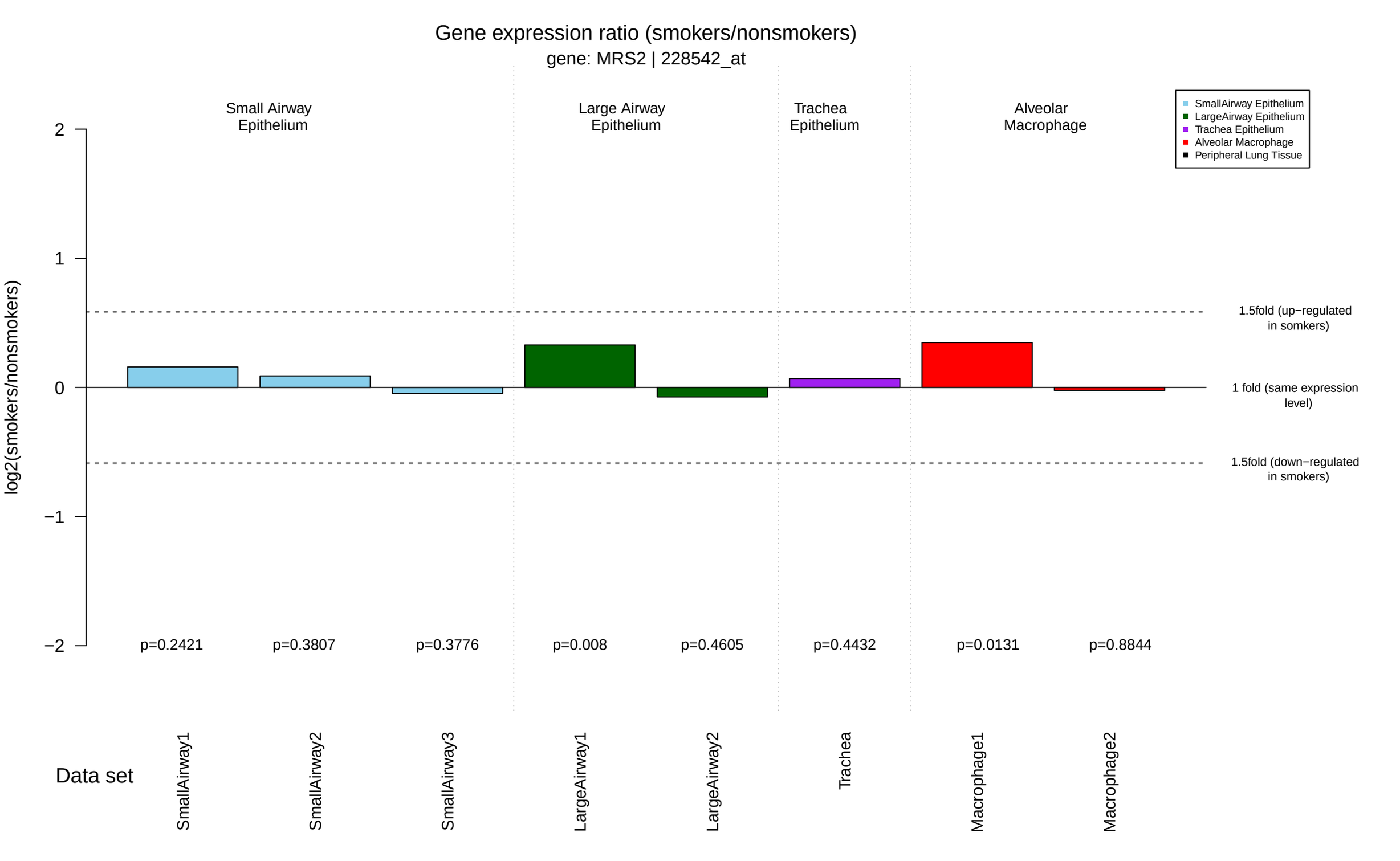
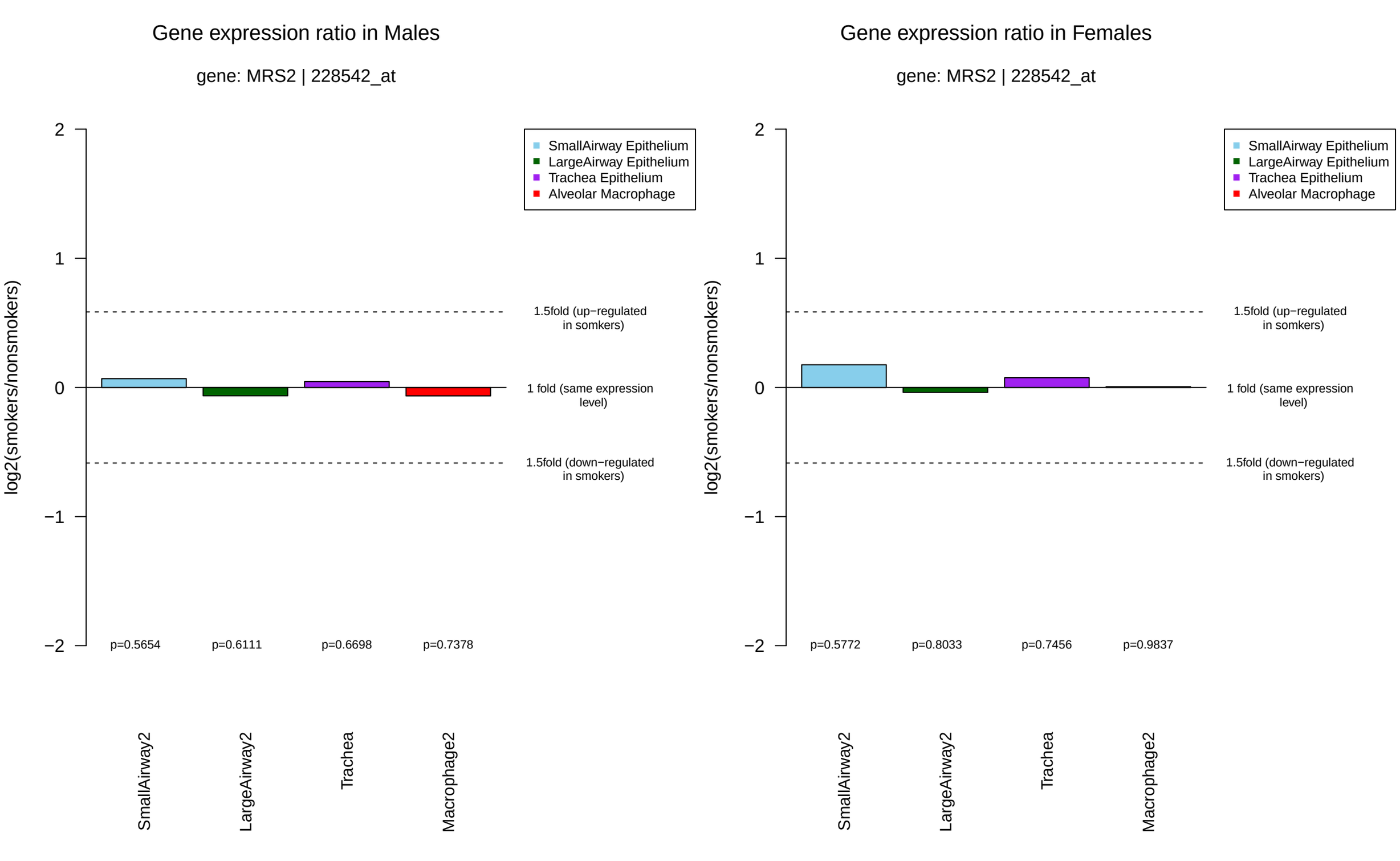
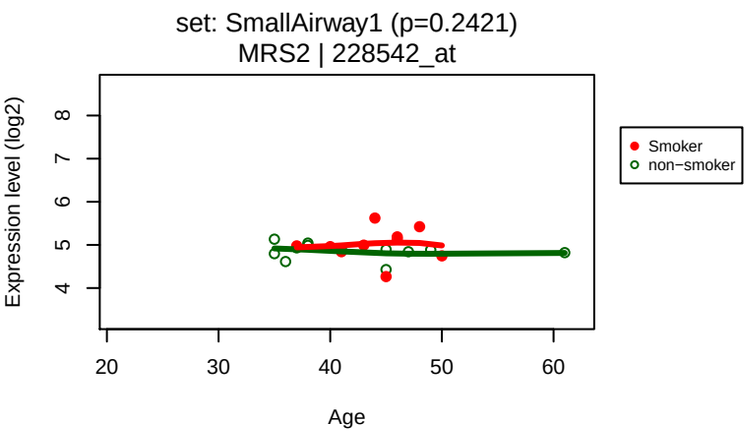
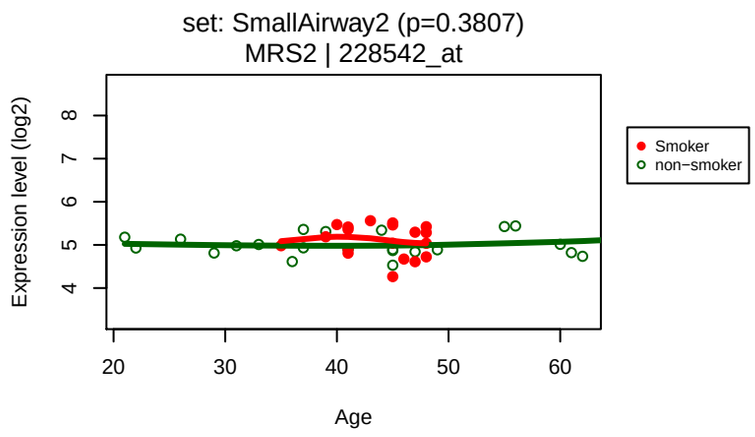
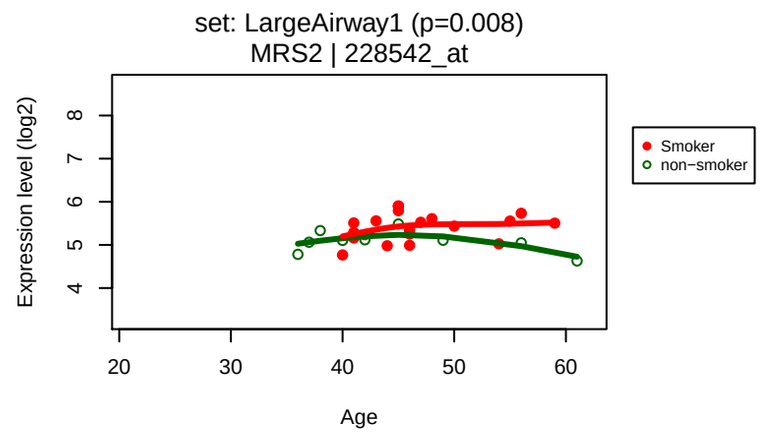

Gene expression of smokers and nonsmokers
gene: MRS2 | 228542_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2421
p=0.3807
p=0.3776
p=0.008
p=0.4605
p=0.4432
p=0.0131
p=0.8844
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

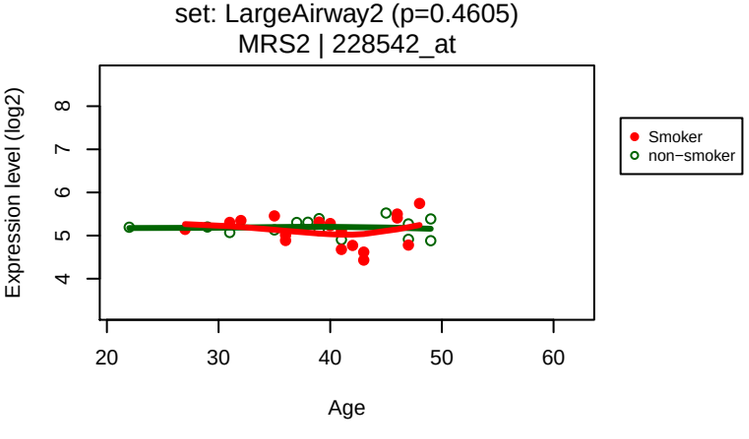

Gene expression ratio (smokers/nonsmokers)
gene: MRS2 | 228542_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2421
p=0.3807
p=0.3776
p=0.008
p=0.4605
p=0.4432
p=0.0131
p=0.8844
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

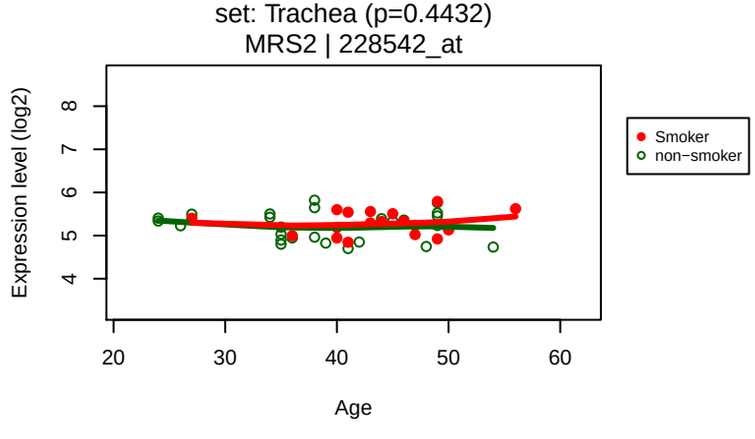

Gene expression ratio in Males
gene: MRS2 | 228542_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5654
p=0.6111
p=0.6698
p=0.7378
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRS2 | 228542_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5772
p=0.8033
p=0.7456
p=0.9837
SmallAirway2
LargeAirway2
Trachea
Macrophage2

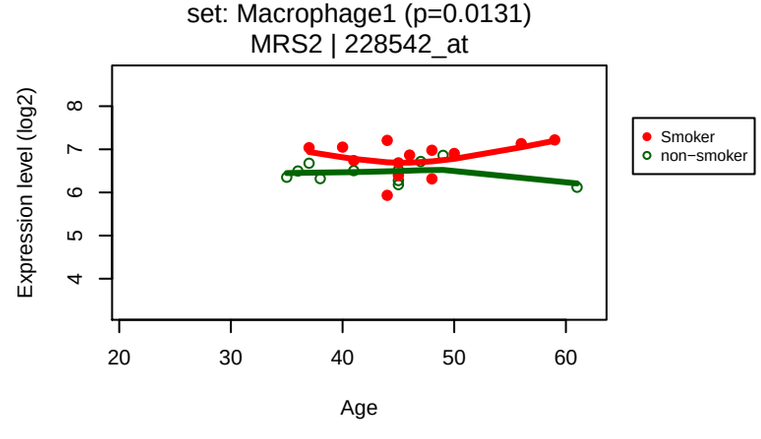

set: SmallAirway1 (p=0.2421)
MRS2 | 228542_at
Expression level (log2)
Age
Smoker
non−smoker

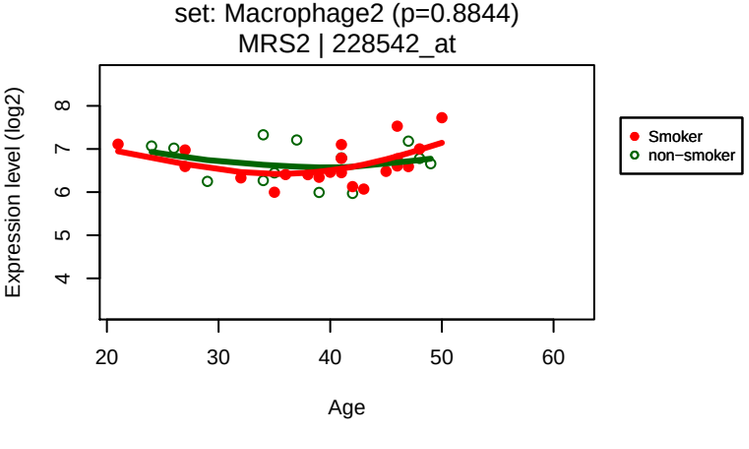

set: SmallAirway2 (p=0.3807)
MRS2 | 228542_at
Expression level (log2)
Age
Smoker
non−smoker

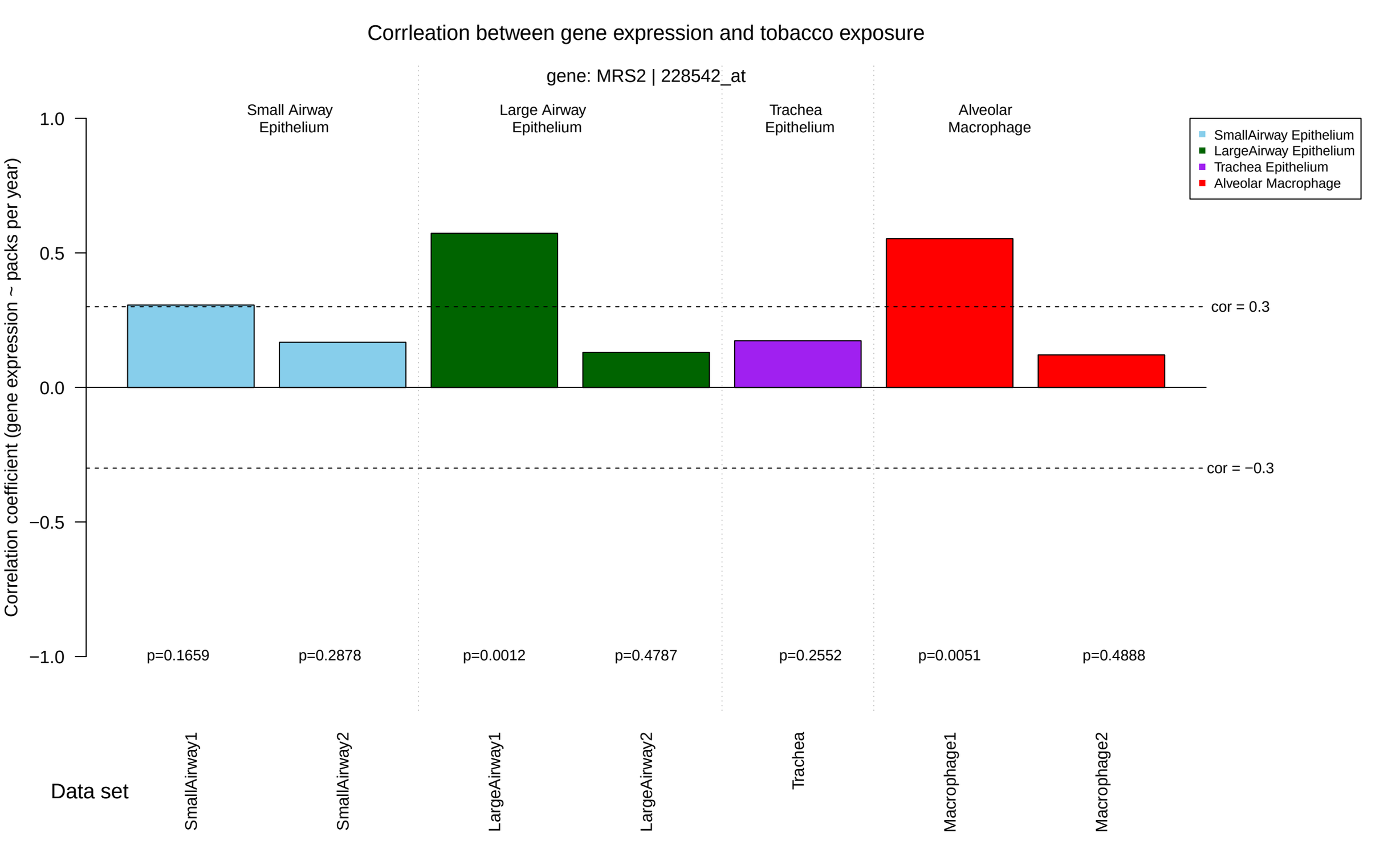

set: LargeAirway1 (p=0.008)
MRS2 | 228542_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4605)
MRS2 | 228542_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4432)
MRS2 | 228542_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0131)
MRS2 | 228542_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8844)
MRS2 | 228542_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRS2 | 228542_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1659
p=0.2878
p=0.0012
p=0.4787
p=0.2552
p=0.0051
p=0.4888
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2